Trader Behavior vs Market Sentiment

Web3 Trading Team – Data Science Assignment

1. Introduction

Market sentiment plays a critical role in shaping trader behaviour, particularly in highly volatile crypto markets. This study analyses how trading activity, profitability, and risk-taking behaviour differ under **Fear** and **Greed** market conditions. By combining historical trader-level data from Hyperliquid with the Bitcoin Fear & Greed Index, the objective is to identify behavioural patterns and inefficiencies that can inform more effective trading strategies.

2. Data & Methodology

Two datasets were used: historical trader data containing detailed trade information (PnL, trade size, direction, and timestamps) and the Bitcoin Fear & Greed Index providing daily market sentiment. Sentiment classifications were consolidated into two regimes—**Fear** and **Greed**—to align with the analysis objective.

Data cleaning included standardizing timestamps, handling missing values, and engineering behavioural features such as trade volume (used as a proxy for risk exposure), profitability indicators, and trade execution hour. The datasets were merged using the trade date as the common key, enabling each trade to be analysed within its corresponding sentiment context.

3. Key Findings

3.1 Trading Activity & Profitability

Trading activity is significantly higher during **Fear** periods, indicating that traders are more active when markets are uncertain or declining. However, despite lower trade counts, **Greed periods deliver higher average profitability per trade**. This suggests that optimism and clearer market trends during Greed allow traders to capture larger gains when trades succeed.

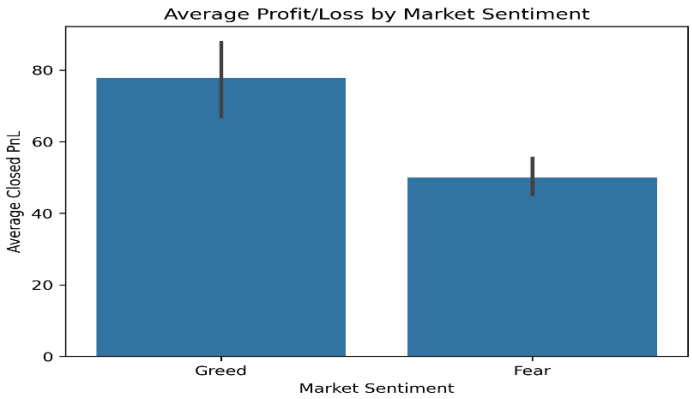

Figure 1: Average Profit/Loss by Market Sentiment

3.2 Win Rate Comparison

Trades executed during Greed exhibit a **higher win rate** compared to Fear. This indicates that traders are more frequently correct during Greed phases, likely due to stronger momentum and reduced ambiguity in market direction. In contrast, Fear periods involve more frequent but less accurate trading.

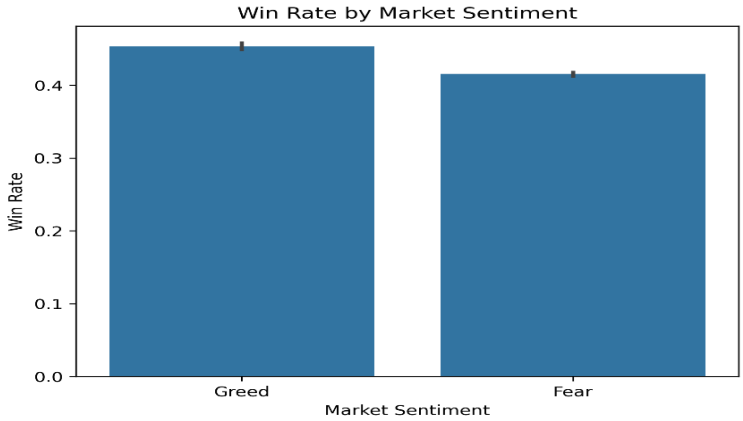

Figure 2: Win Rate by Market Sentiment

3.3 Risk Exposure & Position Sizing

Contrary to intuition, traders assume **larger position sizes during Fear**. Both average trade volume and overall risk exposure are higher in Fear compared to Greed. This suggests that traders often respond to fearful markets with aggressive positioning, potentially attempting to "buy the dip" or anticipate reversals rather than reducing risk.

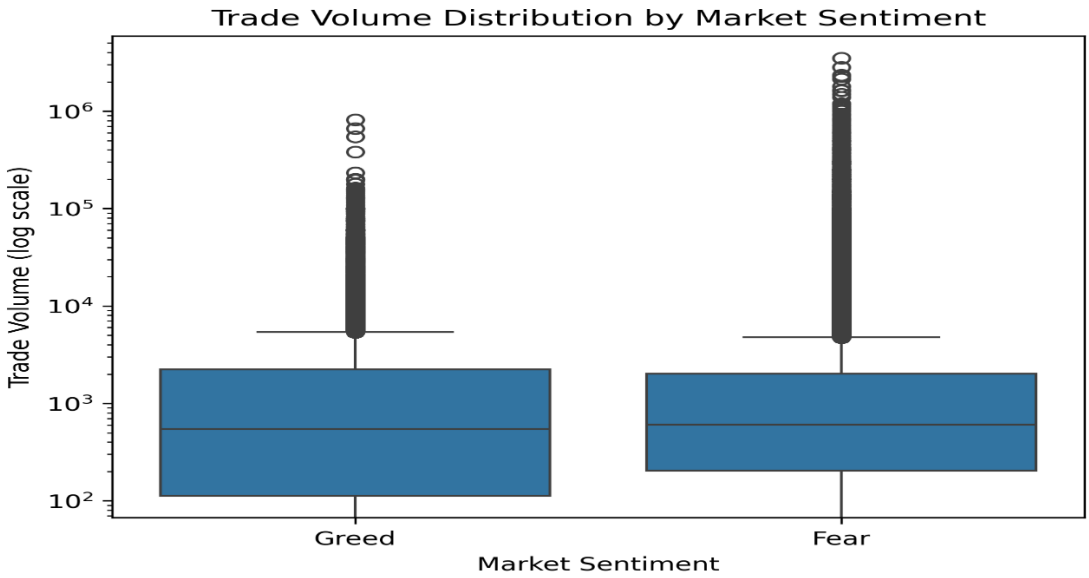

Figure 3: Trade Volume Distribution by Market Sentiment

3.4 Risk-Adjusted Performance

While Fear periods involve higher risk-taking, this risk is **not efficiently rewarded**. Risk-adjusted return analysis shows that **Greed delivers substantially higher profit per unit of risk**. This indicates that risk taken during Greed is more productive, whereas Fear-driven trades suffer from inefficient capital deployment.

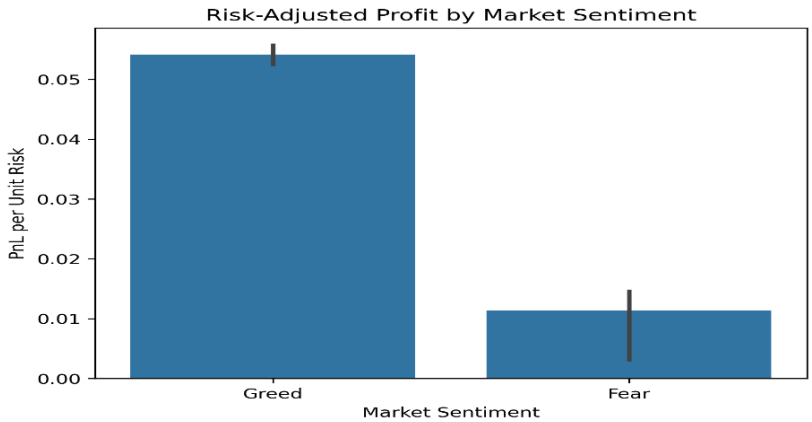

Figure 4: Risk-Adjusted Profit by Market Sentiment

3.5 BUY vs SELL Behaviour

SELL-side trades outperform BUY-side trades across both sentiment regimes. SELL trades demonstrate higher win rates and higher average profitability, particularly during Greed. This suggests that traders are more effective at timing exits or short positions than initiating long entries, especially in optimistic or overheated market conditions.

4. Strategy Implications

The analysis highlights several actionable insights. Increased trading activity and larger position sizes during Fear do not translate into superior performance, indicating potential overconfidence and emotionally driven decision-making. In contrast, Greed environments provide higher profitability, better win rates, and significantly improved risk-adjusted returns despite smaller average position sizes. Traders may benefit from reducing exposure during Fear and focusing on disciplined, momentum-aligned strategies during Greed.

5. Conclusion

This study demonstrates that trader behaviour varies substantially across market sentiment regimes. **Fear periods are characterized by aggressive risk-taking and high activity but poor risk efficiency**, while **Greed periods offer superior profitability and risk-adjusted performance**. The findings suggest that sentiment-aware trading strategies and improved risk management during Fear can meaningfully enhance trading outcomes. These insights provide a strong foundation for developing sentiment-driven decision frameworks in Web3 trading environments.